

RATIONAL BASIS FOR MOMENTUM

One of the first attempts at a risk-based explanation for momentum was by Conrad and Kaul (1998). They postulated that cross-sectional variations in expected returns of individual stocks could account for momentum profits. Jegadeesh and Titman (2001), however, found estimation errors in the work of Conrad and Kaul. Jegadeesh and Titman also argued that post-momentum holding period reversals are not consistent with claims that momentum profits come from variations in expected returns. Grundy and Martin (2001) also showed that expected returns from time-varying risk factors do not explain momentum profitability.

RISK-BASED MOMENTUM MODELS

Chordia and Shivakumar (2002) identified some additional risk factors that they hoped would explain momentum profits. These were lagged macroeconomic variables related to the business cycle. Other explanatory risk factors were stochastic, episodic growth shocks, identified by Johnson (2002). Then came nonparametric, stochastic risks related to industry factors described by Ahn, Conrad, and Dittmar (2003), followed by fluctuations in aggregate liquidity identified by Pastor and Stambaugh (2003). Bansal, Dittmar, and Lundblad (2005) identified consumption risk embedded in cash flows, while Sagi and Seasholes (2007) attributed momentum profits to firm-specific attributes like high market-to-book ratio, high revenue volatility, and low cost of goods sold. Liu and Zhang (2008) linked momentum profits to the growth rate of industrial production.

Countering the use of additional risk factors, Griffin, Ji, and Martin (2003) showed that macroeconomic risk variables do not explain momentum profits. Avramov and Chordia (2006) found evidence challenging time-varying macroeconomic variables and liquidity as explanatory variables of momentum. As we will see in a [Chapter 9](#), data mining and overfitting bias also weigh heavily against the search for additional factors that purport to explain momentum on a rational basis.